

Review?

$$L \in SL_2 \mathbb{Z} \setminus \{ \text{finite order elts in } SL_2 \mathbb{Z} \}$$



↑ finite set

λ_u, λ_s : irrational. ($\lambda_u > 1, 0 < \lambda_s < 1$)

$F_L : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ "hyperbolic toral automorphism"

• $\{ \text{periodic pts of } F_L \} = \text{dense}$

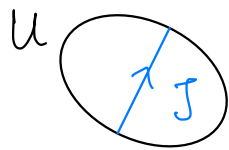
• F_L : topologically transitive

(i.e. $\exists x \in \mathbb{T}^2$ s.t. $\{ F_L^n(x) \}_{n \in \mathbb{Z}} = \text{dense}$)

• $P_n(F_L^n) = \lambda_u^n + \lambda_s^n - 2$.

• F_L : topologically mixing.

$\forall \text{ open } U, V, \exists N > 0$ s.t. $n \geq N \Rightarrow F^n(U) \cap V \neq \emptyset$.



unstable direction

$\forall \varepsilon > 0 \exists N$ s.t. $F_L^N(J)$ meets "every" ε -ball

("locally cpe") topologically transitive $\Leftrightarrow$ topologically mixing.

$$\Omega_N := \{ (\dots, \omega_{-1}, \omega_0, \omega_1, \dots) : \omega_i \in \{0, \dots, N-1\} \}$$

$$d_\lambda(\omega, \omega') = \sum \frac{|\omega_n - \omega'_n|}{\lambda^{|n|}} \quad (\lambda > 1)$$

$\lambda = 10N$: useful to calculate topological entropy.

σ_N : full shift

↳ $\cdot$ {periodic pts of σ_N } : dense in Ω_N

$$\cdot P_n(\sigma_N) = N^n$$

$\cdot \sigma_N$: topologically mixing.

A : 0-1 matrix $N \times N \Rightarrow$ topological Markov chain (TMC)

$$\sigma_A := \sigma_N|_{\Omega_A} : \Omega_A \rightarrow \Omega_A$$

$$\{ \omega = (\omega_n) \in \Omega_N : A_{\omega_n, \omega_{n+1}} = 1 \quad \forall n \in \mathbb{Z} \}.$$

A : transitive (if $\exists m > 0$ s.t. A^m is positive)

$\Rightarrow \cdot \sigma_A$: topologically mixing

$\cdot$ {periodic orbits of σ_A } : dense.

$$\cdot P_n(\sigma_A) = \text{tr}(A^n) = \lambda_{\max}^n + \mu_n \approx \lambda_{\max}^n \quad \& \quad \lambda_{\max} > 1.$$

$$\therefore h(\sigma_A) := \lim_{n \rightarrow \infty} \frac{\log \lambda_{\max}^n}{n} = \log \lambda_{\max} > 0.$$

2.4. Topological classification of expanding maps on a circle.

Def. $f: X \rightarrow X$: **expanding** if $\exists \mu > 1, \varepsilon_0 > 0$ s.t.

$$d(f(x), f(y)) > \mu d(x, y) \quad \forall x \neq y \text{ s.t. } d(x, y) < \varepsilon_0.$$

Prop. $M: \text{cpt}$, $\|D_x f\| > \mu > 1$

$\Rightarrow f$: expanding.

sketch) $d(f(x), f(y)) = \inf_{\gamma} \{ \text{length}(\gamma) : \gamma(0) = f(x), \gamma(1) = f(y) \}$

$$= \int \|\gamma'\|$$

$$= \int \|D_{\tilde{\gamma}} f \tilde{\gamma}'\|$$

$$> \int \mu \cdot \|\tilde{\gamma}'\|$$

$$\geq \mu \cdot d(x, y)$$

$$\left(f \circ \tilde{\gamma} = \gamma, \quad \begin{matrix} \tilde{\gamma}(0) = x \\ \tilde{\gamma}(1) = y \end{matrix} \right)$$

Rmk. $M = S^1$, $f: S^1 \rightarrow S^1 \Rightarrow |f'| \geq 1$?

Lemma. $F: \mathbb{R} \rightarrow \mathbb{R}$: lift of f

$\Rightarrow$ **deg** $f := F(x+1) - F(x) \in \mathbb{Z}$: well-defined.

proof) $G: \mathbb{R} \rightarrow \mathbb{Z}$, $G(x) = F(x+1) - F(x)$

$\Rightarrow G$: conti. $\Rightarrow G = \text{const.}$

$\tilde{F}$: another lift $\Rightarrow \tilde{F} - F: \mathbb{R} \rightarrow \mathbb{Z}$: conti.

$$\Rightarrow \tilde{F}(x+1) - F(x) = F(x+1) - F(x)$$

Lemma. $\text{deg}: C^0(S^1, S^1) \rightarrow \mathbb{Z}$: conti.

$\Rightarrow$ locally constant.

Rmk. f : expanding $\Rightarrow |\text{deg} f| \geq 1$.

$$E_k: x \mapsto kx \pmod{1}.$$

Thm. f : expanding, $\deg f = k$

$\Rightarrow f \sim E_k$ (topologically conjugate)

($\exists$ homeo h s.t. $f = h^{-1} \circ E_k \circ h$).

$d_{C^0}(f, E_k) \downarrow \Rightarrow$ We can choose h s.t. $d_{C^0}(h, id) \downarrow$.

Rmk "deg" gives "topological classification" of expanding maps on S^1 .

Lemma. $f: S^1 \rightarrow S^1$: cont., $\deg f = k$, $|k| \neq 1$

$\Rightarrow \exists F$: lift of f s.t. $\exists$ fixed pt $p \in [-\frac{1}{2}, \frac{1}{2}]$ of F .

$d_{C^0}(f, E_k) \downarrow \Rightarrow |p| \downarrow$

proof) $H := \tilde{F} - id$,

$$H(\frac{1}{2}) = \tilde{F}(\frac{1}{2}) - \frac{1}{2} = \tilde{F}(-\frac{1}{2}) + k - \frac{1}{2} = H(-\frac{1}{2}) + k - 1.$$

$$|H(\frac{1}{2}) - H(-\frac{1}{2})| = |k-1| \geq 1.$$

$\exists$ integer m between $H(-\frac{1}{2})$ & $H(\frac{1}{2})$

$\Rightarrow \exists p \in [-\frac{1}{2}, \frac{1}{2}]$ s.t. $H(p) = m$.

$F := \tilde{F} - m$

$$\Rightarrow F(p) = \tilde{F}(p) - m = H(p) + p - m = p.$$

$d_{C^0}(f, E_k) \downarrow \Rightarrow F(x) - x$: close to $kx - x$.

i.e. $d(0, (k-1)p) \downarrow$

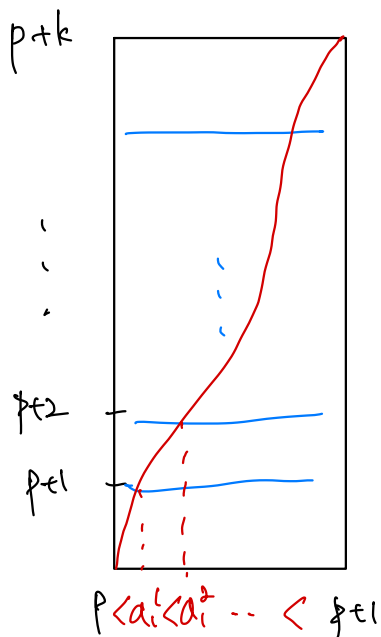
proof of Thm) $k > 1$

$$\text{model} \left\{ \begin{array}{l} \cdot \Delta_0^0 = [0, 1] \\ \cdot \Delta_n^m = \left[\frac{m}{k^n}, \frac{m+1}{k^n} \right] \quad n \in \mathbb{N}, \quad m = 0, \dots, k^n - 1. \\ \cdot \xi_n = \{ \Delta_n^0, \dots, \Delta_n^{k^n-1} \} : \text{partition of } [0, 1]. \\ E_k(\Delta_n^m) = \Delta_{n-1}^{m'} \quad \text{where } m' \equiv m \pmod{k^{n-1}}. \\ \left(\because E_k \left(\left[\frac{m}{k^n}, \frac{m+1}{k^n} \right] \right) = \left[\frac{m}{k^{n-1}}, \frac{m+1}{k^{n-1}} \right] \pmod{1} \right) \end{array} \right.$$

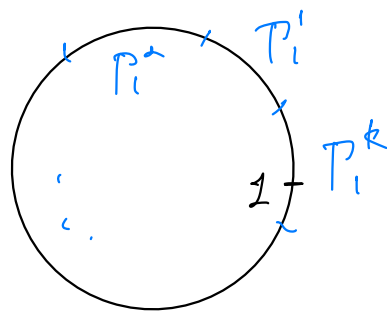
Lemma $\Rightarrow$ Choose a lift F s.t. $F(p) = p$, $F(p+1) = p+k$.

f : expanding $\Rightarrow F$: strictly increasing.

$\Rightarrow a_i^0 = p < a_i^1 < \dots < a_i^{k-1} < a_i^k = p+1$: partition
s.t. $F(a_i^m) = p+m$.



$$T_1^m := \pi([a_i^m, a_i^{m+1}]) \subset S'.$$



$$f(T_1^m) = S'.$$

* f : close to $E_k \Rightarrow p$: close to 0

a_i^m : close to $\frac{m}{k}$.

In $[a_1^m, a_1^{m+1}]$, choose $a_2^{km}, \dots, a_2^{km+k}$ s.t.

$$F(a_2^{km+k}) = p+m + \frac{k}{k}.$$

$$T_2^{\tilde{p}} := \pi([a_2^{\tilde{p}}, a_2^{\tilde{p}+1}]) \Rightarrow f(T_2^{\tilde{p}}) = T_1^{m'} \subset S^1$$

$\vdots$

$$\Rightarrow a_n^0, \dots, a_n^{k^n}, \quad \gamma_n = \{T_n^0, \dots, T_n^{k^n-1}\}.$$

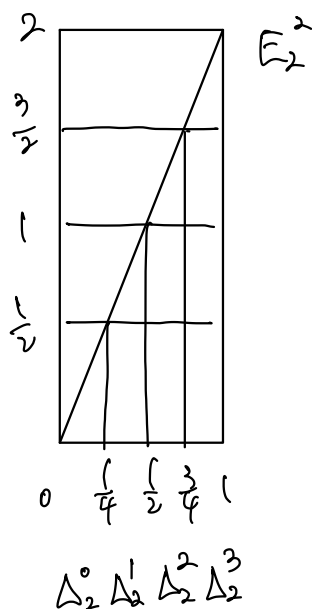
$\updownarrow$

$\sum_n$

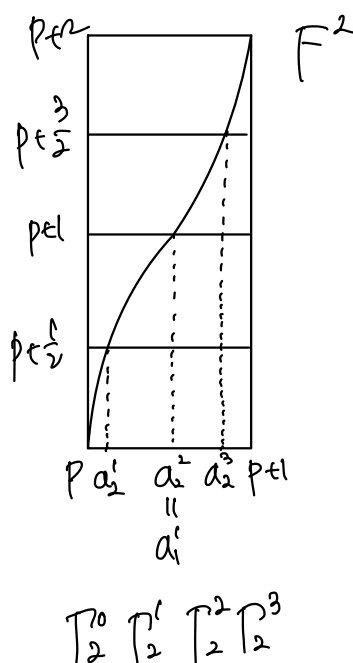
$$\cdot f(T_n^m) = T_n^{m'}, \quad f^n(T_n^m) = S^1.$$

$$\cdot f : \text{close to } E_k \Rightarrow |a_n^m - \frac{m}{k^n}| < \varepsilon/3. \quad (\Rightarrow h : \text{close to id.})$$

$k=2, n=2$



$\Leftrightarrow$



$$|f'| \geq \mu > 1 \Rightarrow \text{length } T_n^m < \mu^{-n}.$$

$$\Rightarrow \{a_n^m\} : \text{dense}.$$

$$\tilde{h}: a_n^m \mapsto \frac{m}{k^n} \quad : \text{strictly monotone}$$

$$\tilde{h} \text{ extends to } h: [p, p+1] \rightarrow [0, 1] \quad \leftarrow$$

$$\Rightarrow h \approx \text{homeo.}$$

$$E_k \circ h = h \circ f \quad \text{on dense set !!}$$

$$\left(\because E_k(h(a_n^m)) = E_k\left(\frac{m}{k^n}\right) = \frac{m'}{k^{n-1}} = h(a_{n-1}^{m'}) = h(f(a_n^m)) \right)$$

$$\therefore f \sim E_k.$$

another proof)

"general methods used in the theory of hyperbolic dynamical systems"

$$\mathcal{C} := \{ h: [0, 1] \rightarrow [0, 1] : \text{continuous}, h(0)=0, h(1)=1 \}$$

$$\mathcal{F}: \mathcal{C} \rightarrow \mathcal{C}$$

$$(\mathcal{F}h)(x) = \begin{cases} \frac{1}{k} h(\{F(x)\}) + \frac{m}{k} \pmod{1}, & x \in T_1^m \\ 1, & x=1 \end{cases}$$

$\frac{m}{k} \sim \frac{m+1}{k}$

$$\Rightarrow \text{dist}(\mathcal{F}h_1, \mathcal{F}h_2) = \sup \left| \frac{1}{k} (h_1(\{F(x)\}) - h_2(\{F(x)\})) \right|$$

$$= \frac{1}{|k|} \text{dist}(h_1, h_2)$$

$$\Rightarrow \mathcal{F}: \text{contraction} \Rightarrow \exists h_0 \text{ s.t. } \mathcal{F}h = h_0$$

(Banach fixed point theorem).

$$E_k \circ h_0 = h_0 \circ f$$

$$\tilde{\mathcal{F}}(h) = f_m^{-1}(h(kx)) \quad \left(x \in \left[\frac{m}{k}, \frac{m+1}{k} \right] \right).$$

$$\uparrow \text{branch of } f \Rightarrow \exists \text{ fixed pt } h_1 \text{ \& } h_1 = h_0^{-1}.$$

(11.12.1)

$$h_0 \circ h_1 \circ E_k = h_0 \circ f \circ h_1 = E_k \circ h_0 \circ h_1, \quad \deg h_0 \circ h_1 = 1 \Rightarrow h_0 \circ h_1 = \text{id.}$$

2.5. Coding, horseshoes, and Markov partitions.

$f: X \rightarrow X$, $(X_0, \dots, X_{p-1})$: decomposition of X .

$x \in X \rightsquigarrow (w_n := i \text{ s.t. } f^n(x) \in X_i)_{n \in \mathbb{Z}}$.
coding

$\rightsquigarrow$ Want: $f \circ h = h \circ \theta_N$.

$\left\{ \begin{array}{l} \cdot X_i \cap X_j = \emptyset ? \quad (\Delta_n^m \cap \Delta_n^{m+1} = pt) \\ \cdot \left| \bigcap_{n \in \mathbb{Z}} f^{-n}(X_{w_n}) \right| \geq 1 \rightsquigarrow \text{same code.} \end{array} \right.$ well-defined?

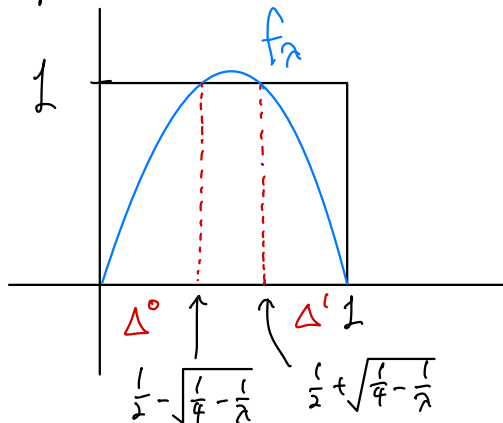
Markov partition: "Good" decomposition $(X_0, \dots, X_{p-1})$

$\Rightarrow$ semiconjugacy h between TMC θ_A & f .
 (surj.)

1-1 on "large set" & Markov way.

* quadratic map.

$f_\lambda: \mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto \lambda x(1-x)$. ($\lambda \geq 2 + \sqrt{5}$)

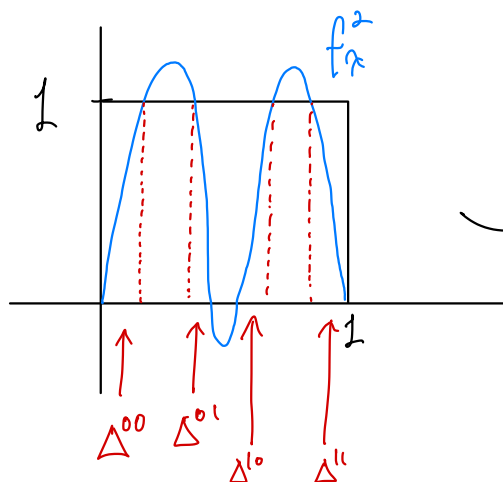


$$|f'_\lambda| \geq \sqrt{\lambda^2 - 4\lambda} > 1 \text{ on } \Delta^0 \cup \Delta^1$$

$$\lim_{n \rightarrow \infty} \bigcap_{k=0}^n f_\lambda^{-k}(\Delta^{w_n}) \downarrow \emptyset$$

$$\bigcap_{n \in \mathbb{Z}} f_\lambda^{-n}(\Delta^{w_n}) = pt$$

$$f_\lambda^{-1}([0, 1]) = \Delta^0 \cup \Delta^1$$



$$f_\lambda^{-2}([0, 1]) = \Delta^{00} \cup \Delta^{01} \cup \Delta^{10} \cup \Delta^{11}$$

$$\cdot \Lambda = \bigcap_{n \in \mathbb{N}} f_n^{-n}([0,1]) : \text{Cantor set.}$$

$$\cdot h: \Omega_2^R \rightarrow \Lambda : \text{homeo.}$$

$$w \mapsto \bigcap_{n \geq 0} f_n^{-n}(\Delta^{w_n})$$

$$h \circ \sigma_2^R = f_n|_{\Lambda} \circ h$$

one-sided 2-shift $\Leftrightarrow$ dynamics on Λ .

* horseshoe

$\Delta \subset \mathbb{R}^2$: rectangle.

$f: \Delta \rightarrow \mathbb{R}^2$: "horseshoe" diffeo.

$$\Delta \cap f(\Delta) = \Delta_0 \cup \Delta_1$$

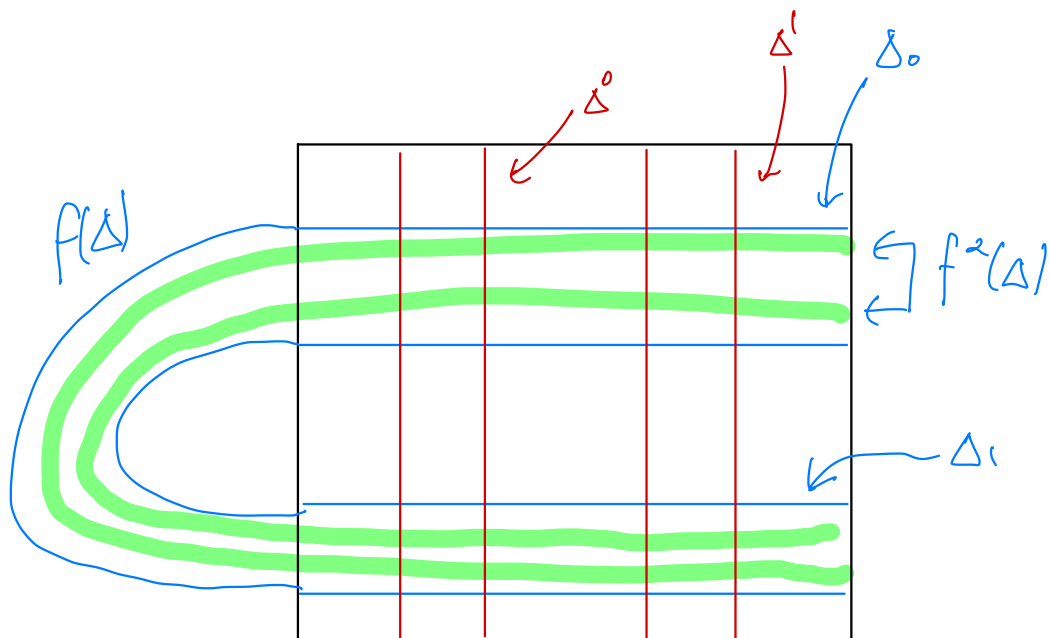


horizontal rectangles.

$$\Delta \cap f^{-1}(\Delta) = \Delta^0 \cup \Delta^1$$



vertical rectangles.



$$\cdot \Lambda = \bigcap_{n \in \mathbb{Z}} f^n(\Delta) = \text{Cantor set} \times \text{Cantor set}.$$

$$\cdot h: \Omega_2 \rightarrow \Lambda$$

$$\omega \mapsto \bigcap_{n \in \mathbb{Z}} f^{-n}(\Delta^{\omega_n})$$

$$\cdot h \circ \sigma_2 = f|_{\Lambda} \circ h$$

Cor. $\cdot \{\text{periodic pts of } f|_{\Lambda}\}$ dense in Λ .

$$\cdot P_n(f|_{\Lambda}) = 2^n$$

$\cdot f|_{\Lambda}$: topologically mixing.

Further $\cdot \Delta^0, \dots, \Delta^{N-1} \leftrightarrow 0-1 \text{ matrix TMC}.$

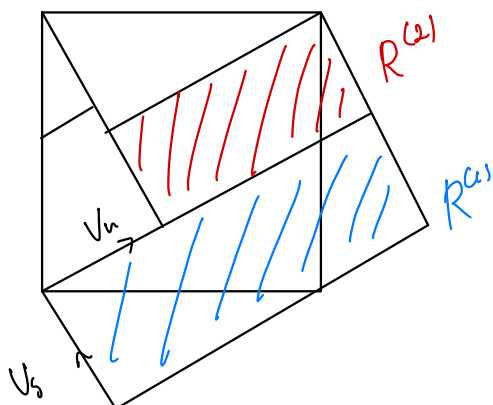
$$f^{-1}(\Delta^i) \cap \Delta^j \leftrightarrow a_{ij} = 0 \text{ or } 1$$

$\cdot$ higher dim horseshoe

$\cdot$ nonlinear horseshoe (perturbation $\Rightarrow$?)

* hyperbolic toral automorphism.

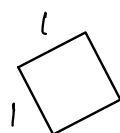
$$F(x, y) = F_L(x, y) = (2x + y, x + y) \pmod{1}$$



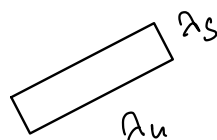
$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \Rightarrow \lambda_u = \frac{3+\sqrt{5}}{2}$$

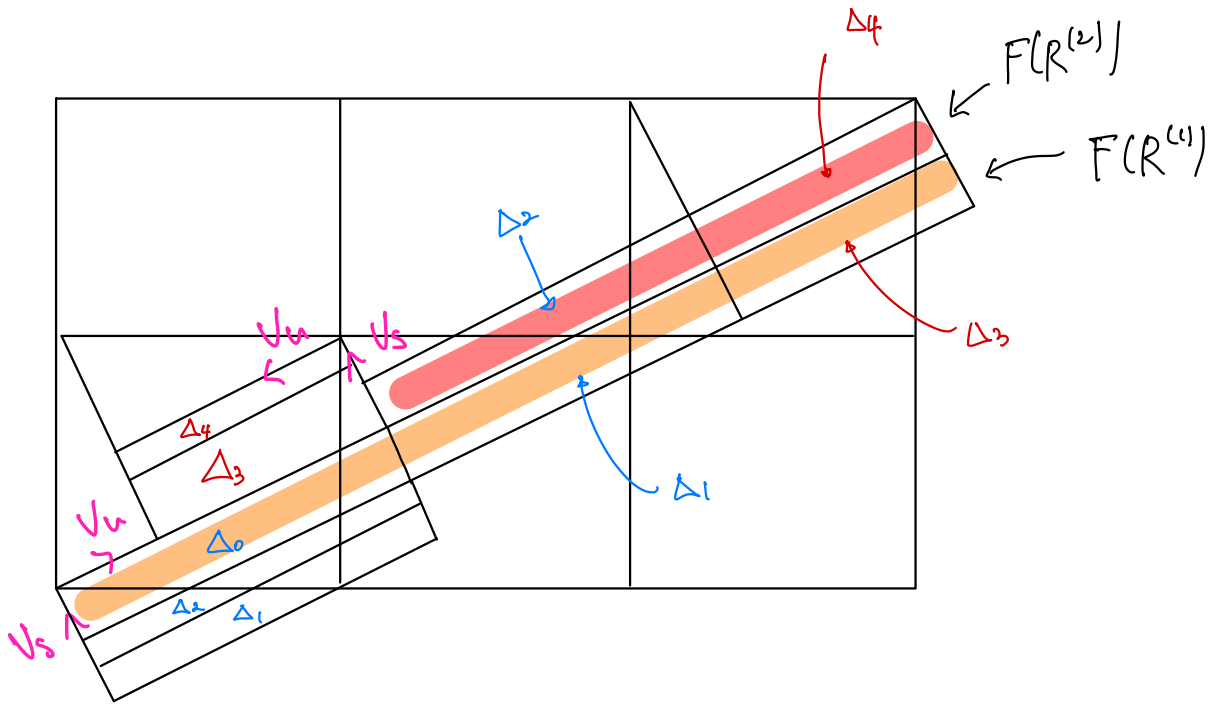
$$\lambda_s = \frac{3-\sqrt{5}}{2}$$

$$\Downarrow$$



$\rightsquigarrow$





F < contracting in "vertical" direction
expanding in "horizontal" direction

$$x \in \Delta_i \Rightarrow W_0 = i.$$

$$F(\Delta_i) \cap \Delta_j \neq \emptyset \text{ for some } j.$$

$$W_n \text{'s satisfy } F(\Delta_{W_n}) \cap F(\Delta_{W_{n+1}}) \neq \emptyset.$$

$\leadsto$ "Markov" property

$$\begin{cases} F(\Delta_0 \cup \Delta_1 \cup \Delta_2) = \Delta_0 \cup \Delta_1 \cup \Delta_3 \\ F(\Delta_3 \cup \Delta_4) = \Delta_2 \cup \Delta_4 \end{cases}$$

$$\Rightarrow A = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

F : contracting in "vertical" direction

F^{-1} : "horizontal" "

$$\left| \bigcap_{n \in \mathbb{Z}} F^n(\Delta_{\omega_n}) \right| \leq 1 \quad \forall \omega = (\omega_n) \in \Omega_5.$$

$$h: \Omega_A \rightarrow \mathbb{T}^2$$

$\parallel$

$$\{ \omega \in \Omega_5 : A_{\omega_n, \omega_{n+1}} = 1 \quad \forall n \in \mathbb{Z} \}.$$

semiconjugacy $\rightarrow F \circ h = h \circ \sigma_A$

$$\therefore \underbrace{(\omega_n^0 = 0)_{n \in \mathbb{Z}}, (\omega_n^1 = 1)_{n \in \mathbb{Z}}, (\omega_n^4 = 4)_{n \in \mathbb{Z}}}_{\text{Fixed seq. in } \Omega_A}$$

$$\text{But, } F(x,y) = (x,y) \Leftrightarrow (x,y) = (0,0).$$

$$\begin{cases} P_n(F_L) = \lambda_u^n + \lambda_s^n - 2 \\ P_n(\sigma_A) = \text{tr}(A^n) = \lambda_u^n + \lambda_s^n \quad (\lambda \text{ of } A = \lambda_u, \lambda_s, 3 \cdot 0) \\ \text{i.e. } P_n(\sigma_A) = P_n(F_L) + 2 \end{cases}$$

$z \in \mathbb{T}^2$ s.t. pos/neg iterates avoid $\partial R^{(1)}, \partial R^{(2)}$.

$$\Leftrightarrow \exists! \text{ preimage.}$$

Cor. $\cdot h: 1-1$ on all periodic pts except fixed pts.

$\cdot |h^{-1}(z)| < \infty$ if $F^{-n}(z) \neq 0$

proof) $\cdot F^m(x, y) = (x, y) \Leftrightarrow \left(\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^m - I \right) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\Rightarrow (x, y) \in \mathbb{Q}.$$

$$\Rightarrow F^n(x, y) \notin 2\mathbb{R}^{(1)} \cup 2\mathbb{R}^{(2)} \\ (\text{except } (x, y) = (0, 0))$$

$$\Rightarrow \exists! \omega_0, \dots, \omega_{m-1} \text{ s.t.}$$

$$(x, y) \in \Delta_{\omega_0}, \dots, F^{m-1}(x, y) \in \Delta_{\omega_{m-1}}$$

$\cdot$ Assume $F^{-n}(z) \neq 0$. $z \notin W^u(0)$.

Choose $\omega \in h^{-1}(z)$.

$$F^{-n}(z) = h(\sigma_A^{-n}(\omega)).$$

ω_n : uniquely determined

$$\therefore |h^{-1}(z)| \leq 3.$$

$$z \in W^s(0) \Rightarrow 0\text{'s or } 1\text{'s or } 4\text{'s}.$$

Rmk. $F^n(z) \rightarrow 0$ & $\omega \in h^{-1}(z)$

$$\begin{cases} \forall n \geq N, \omega_n = 0, \text{ or} \\ \forall n \geq N, \omega_n = 1, \text{ or} \\ \forall n \geq N, \omega_n = 4 \end{cases}$$

i.e. finitely many preimage.